

GAGAN JAIN

Software Engineer · Full Stack & AI/LLM Developer

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SUMMARY

Computer Science student (B.Tech, VIT Vellore, expected July 2026, CGPA 7.5+) with hands-on project experience across full-stack web development, developer tooling, and systems programming. Designed and implemented a custom programming language with a complete compiler pipeline (lexer, parser, type checker, bytecode compiler, VM) and a browser-based IDE. Built a production-hardened Flask application with Argon2id auth, CI/CD, Docker deployment, and monitoring. Completed an Industry 5.0 industrial automation training internship. Seeking entry-level software engineering roles.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, Rust, HTML, CSS

Frontend: React, React Native, Tailwind CSS, Bootstrap

Backend: Node.js, Express, Flask, Axum (Rust), REST APIs

Databases: MySQL, MongoDB Atlas

DevOps & Tools: Git, Docker, VS Code, Monaco Editor, Linux

AI/ML & Other: YOLO, Computer Vision, NLP, Ollama, IoT (Raspberry Pi, Arduino)

PROJECTS

Vyākṛti — Programming Language & Web-Based IDE

[Rust](#), [React](#), [TypeScript](#), [Monaco Editor](#), [Zustand](#), [Tailwind CSS](#), [Axum](#)

- Designed and implemented a custom Sanskrit-oriented programming language: 38+ token type lexer, recursive-descent parser, kāraka-driven semantic type checker, bytecode compiler (34 opcodes), stack-based VM
- Built an integrated browser-based IDE with React and Monaco Editor supporting code editing, compilation, and interactive REPL via WebSocket
- Engineered a Rust (axum) backend providing compile, REPL, LSP, and file-management endpoints
- 123 tests covering the full pipeline; self-hosting corpus with language parsing its own source

AIM — Attendance Information Manager

[Python](#), [Flask](#), [MySQL](#), [Bootstrap](#), [Chart.js](#), [Docker](#), [GitHub Actions](#)

- Built a production-hardened full-stack attendance management app with layered architecture (routes → services → repositories → models)
- Implemented Argon2id password hashing, CSRF protection, brute-force lockout, breached-password detection, rate limiting, and security headers (CSP/HSTS)
- Added reporting with Chart.js visualizations, FullCalendar integration, responsive light/dark themes, CSV import/export, encrypted backup/restore

- Configured Prometheus monitoring, structured JSON logging, 84 tests, Docker Compose deployment, and GitHub Actions CI/CD

Food Delivery App

React Native, Node.js, MongoDB Atlas

- Built a cross-platform mobile app with authentication, restaurant browsing, ordering, and order tracking on a Node.js + MongoDB backend

Selected Additional Projects

Sentiment Analysis using Facial Recognition (YOLO/CV) · IoT House Security (Raspberry Pi) · RFID Access Control with Behavioral Intelligence (Arduino + Python ML) · File Integrity Checking System (checksums/hashing)

EXPERIENCE

Industry 5.0 Industrial Automation Trainee (Training Internship)

Aug – Nov 2025

CodenPlay Robotics · VIT Vellore / Online · Mentor: Dr. Konguvel E

- Built and debugged PLC automation logic using ladder logic and structured text in CODESYS
- Simulated industrial process workflows in Factory I/O and implemented flow-based control/integration in Node-RED
- Covered HMI design concepts, industrial communication protocols, process monitoring, and the debugging–testing–deployment workflow

EDUCATION

B.Tech, Computer Science — VIT Vellore · CGPA 7.5+

Expected July 2026